Molecular Symmetry

Why is symmetry important?

Places you have already meet symmetry:

- o in determining the equivalent H or C atoms in an NMR spectrum
- o in deciding if a molecule is chiral
- o in the labelling of atomic orbitals
- o in distinguishing between s and p orbitals in organic chemistry
- o in octahedral transition-metal complexes
- o in determining isomers: cis/trans, fac/mer, staggered/eclipsed

Why is symmetry important?

• the symmetry of a molecule is crucially important to its behavior

o it is used to construct the MO diagram

o it determines the HOMO and LUMO and thus the reactivity of the species

o it determines the vibrations that the molecule can undergo and thus the IR and Raman spectrum

o it determines the electronic interactions the molecule can undergo, thus determining features like the dipole moment, and UV spectrum

Symmetry

Symmetry

Symmetry is all around us and is a fundamental property of nature

The idea of symmetry is a familiar one

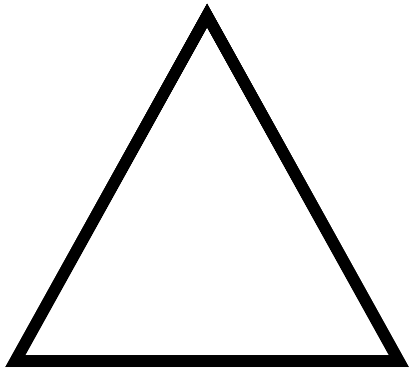
we speak of a shape as being symmetrical , unsymmetrical or even more symmetrical than some other shape .

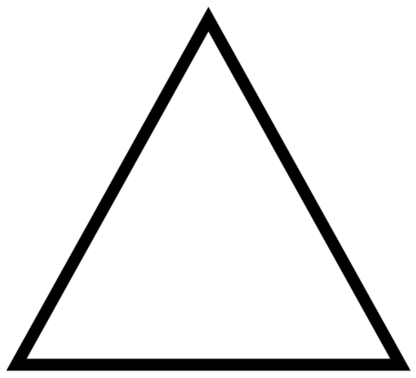
For scientific purposes, however, we need to specify ideas of symmetry in a more quantitative way.

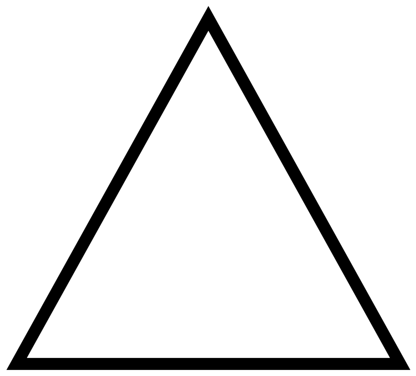
Symmetry Element

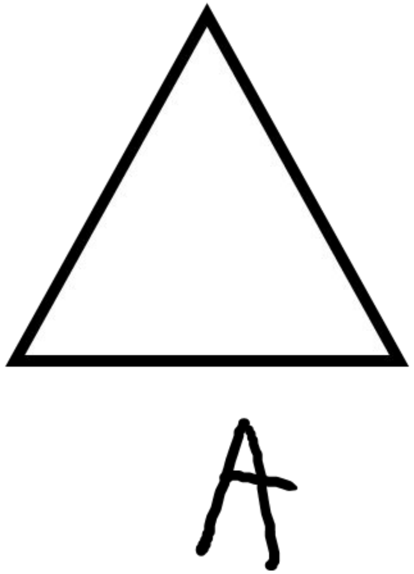
Symmetry Operation

Point group

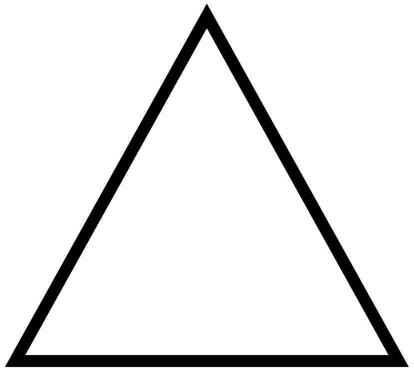


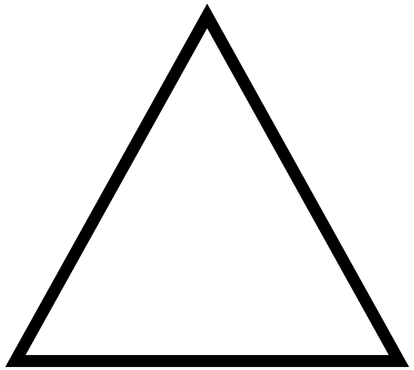






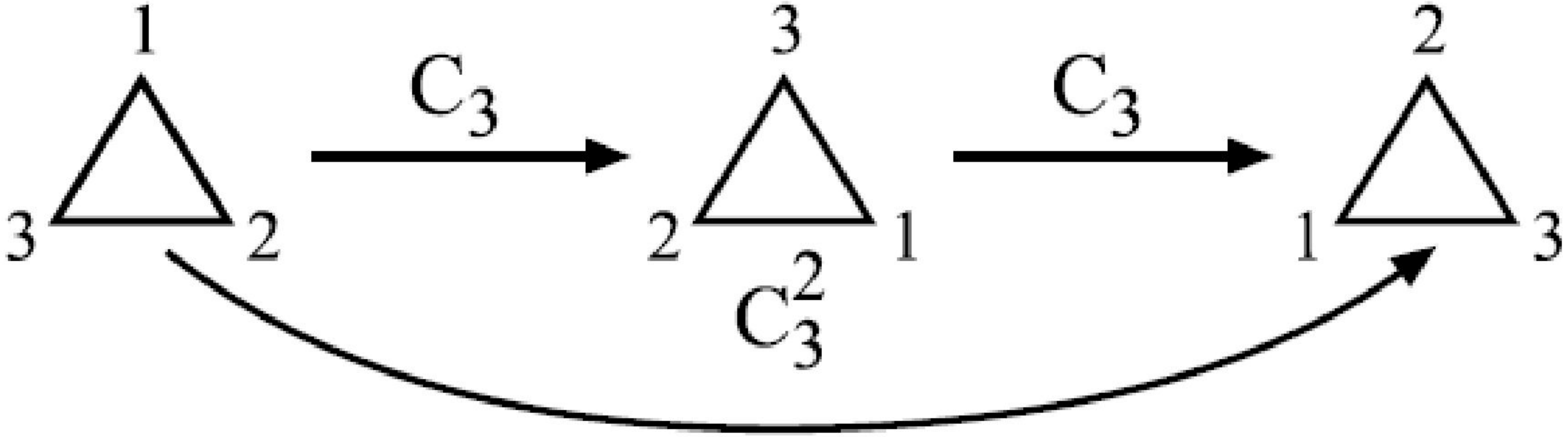
Can you think of another operation you could perform on a triangle which is also a symmetry operation?

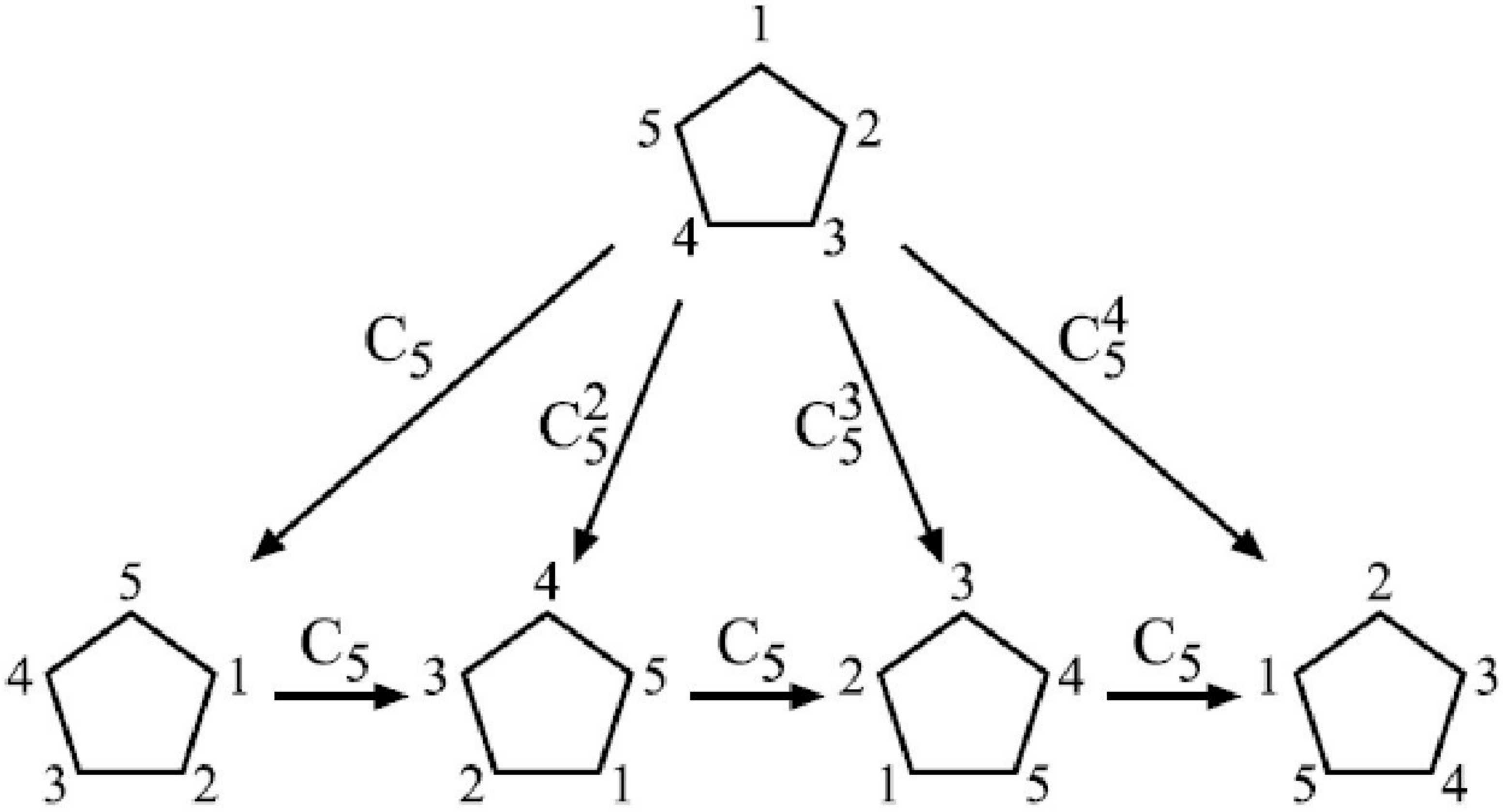




A symmetry operation is the operation of actually doing something to a shape so that the result is indistinguishable from the initial state.

One element of symmetry may generate more than one operation



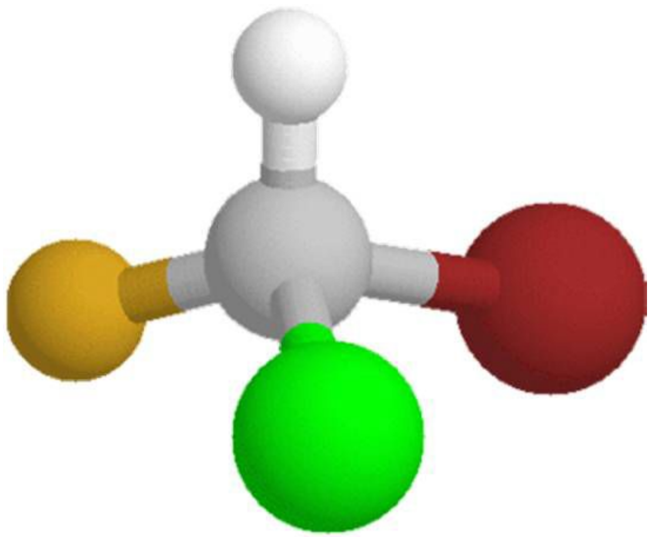


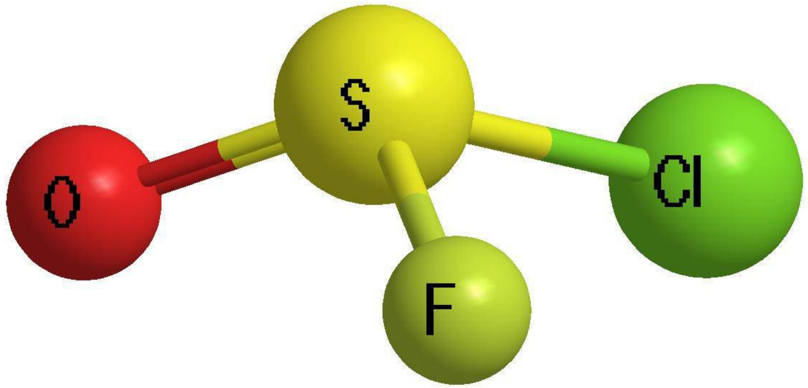
The shape is now more than indistinguishable, it is IDENTICAL with the starting point.
We say that

where E is the IDENTITY OPERATION, or the operation of doing nothing.

Identity (E)

Simplest symmetry operation. All molecules have this element. If the molecule does have no other elements, it is asymmetric

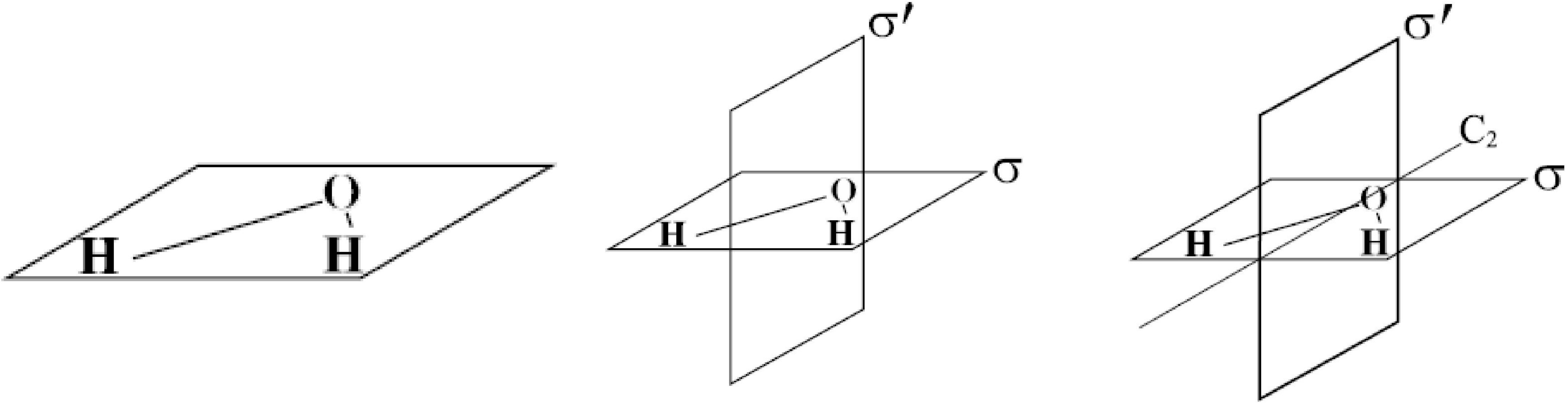


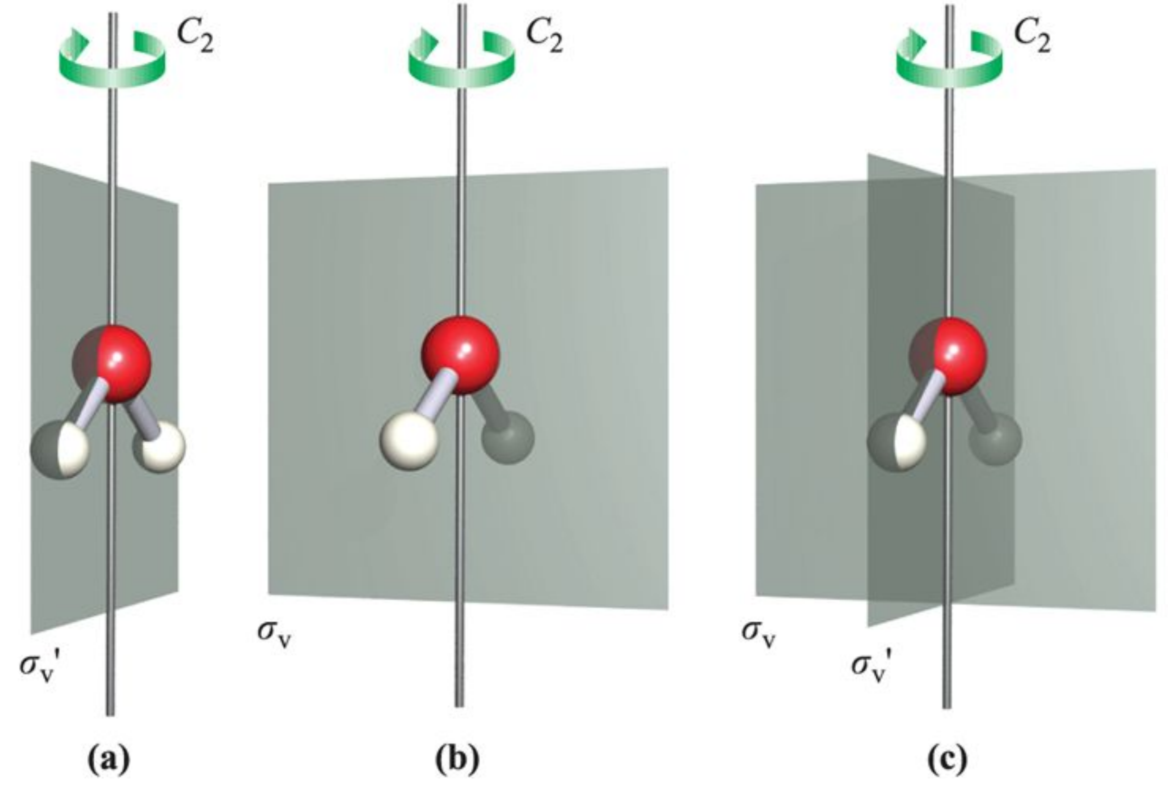


A plane of symmetry:

This is given the symbol σ (Sigma)

The element generates only one operation, that of reflection in the plane.



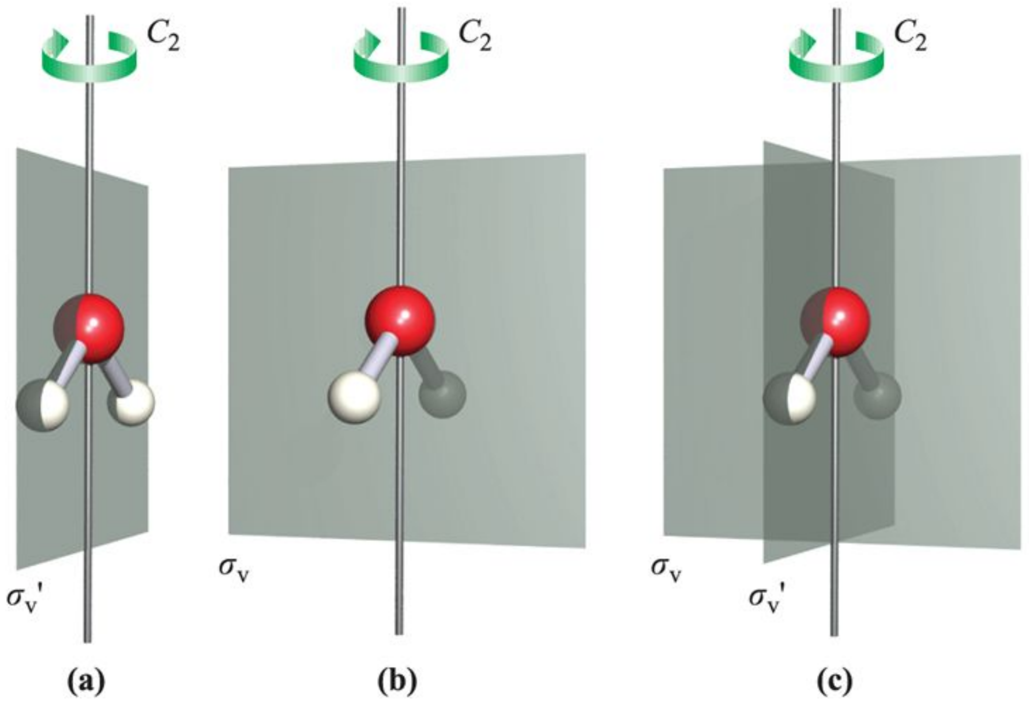


If reflection of all parts of a molecule through a plane produced an indistinguishable configuration, the symmetry element is called a mirror plane or plane of symmetry.

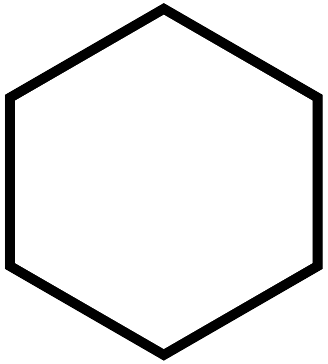
σv (vertical): plane colinear with principal axis

σh (Horizontal): Plane perpendicular to principal axis

σd (dihedral): Vertical, parallel to principal axis, σ parallel to Cn and bisecting two C2' axes

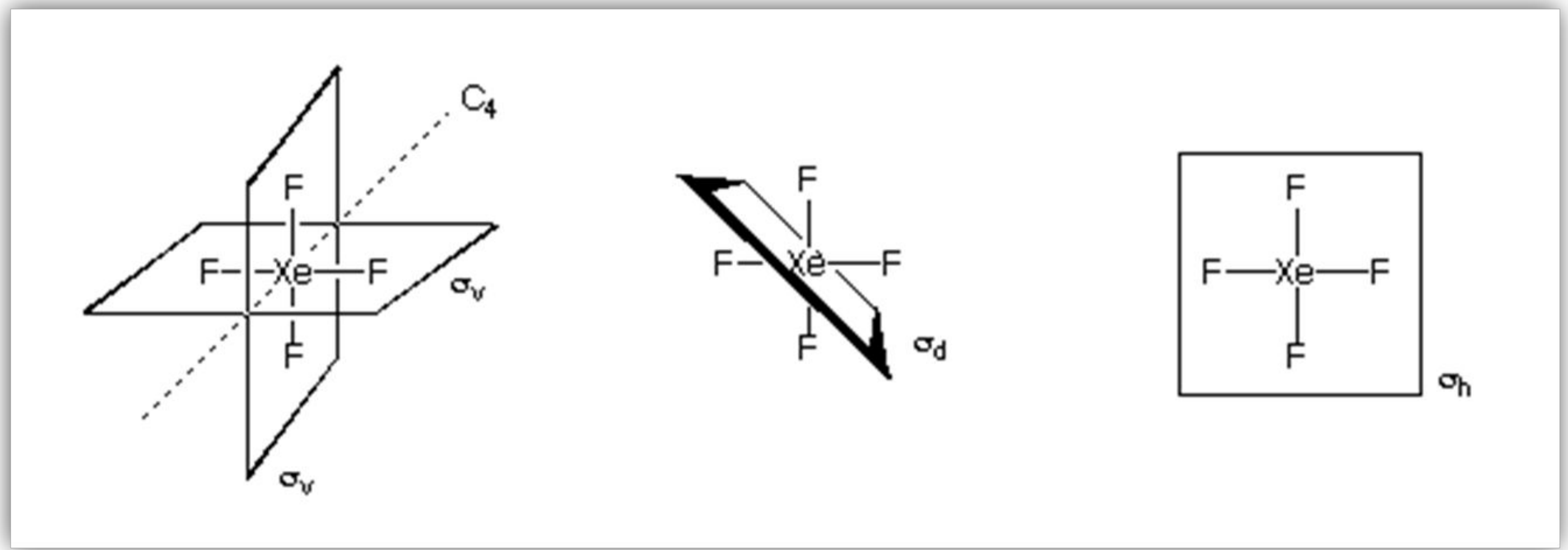


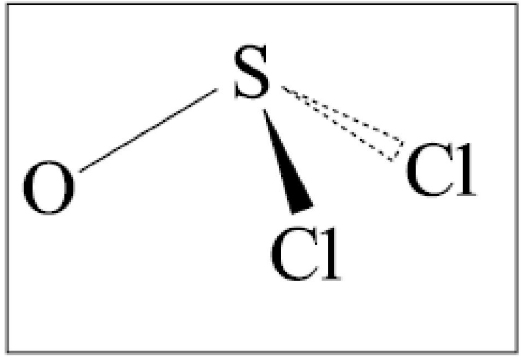
This completes the description of the symmetry of water. It actually has FOUR elements of symmetry



Many molecules have this set of symmetry elements, so it is convenient to classify them all under one name,

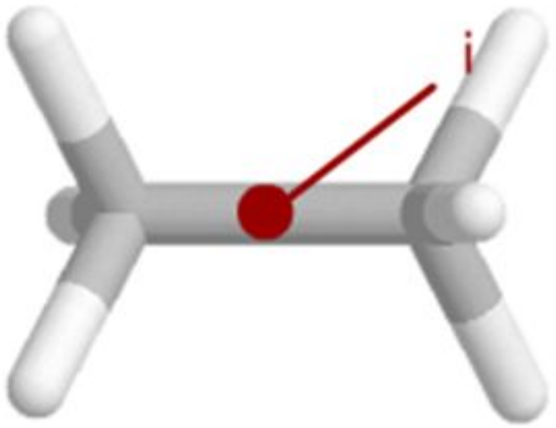
The set of such symmetry operations is called the point group


C4
σv
σv
F
F
F
F
Xe
σd
σh

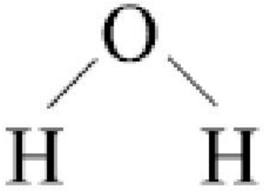

S
O
Cl
Cl

A further element of symmetry is the INVERSION CENTRE, i.
This generates the operation of inversion through the centre.

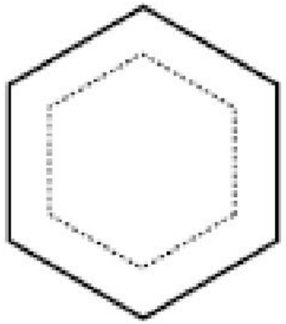
Draw a line from any point to the centre of the molecule, and produce it an equal distance the other side. If it comes to an equivalent point, the operation of inversion is a symmetrynoperation.



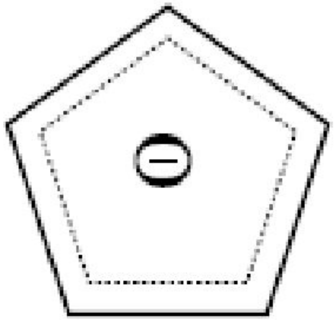
A center of symmetry: A point at the center of the molecule.
(x,y,z)   (-x,-y,-z).



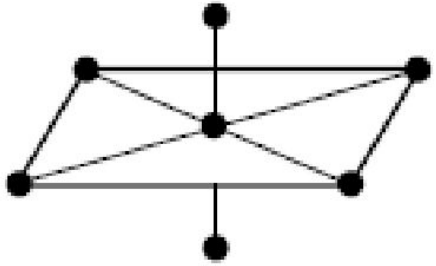
A



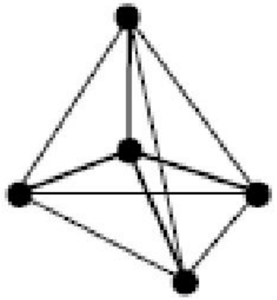
B



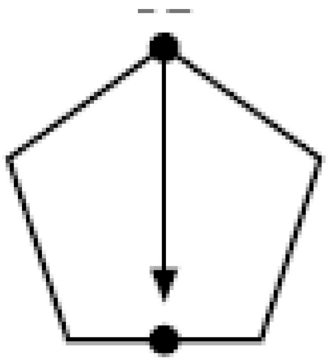
C

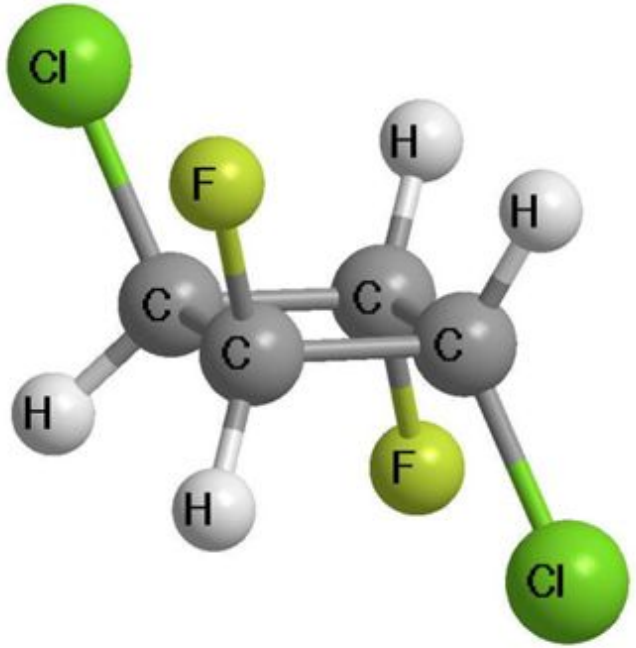


D



E

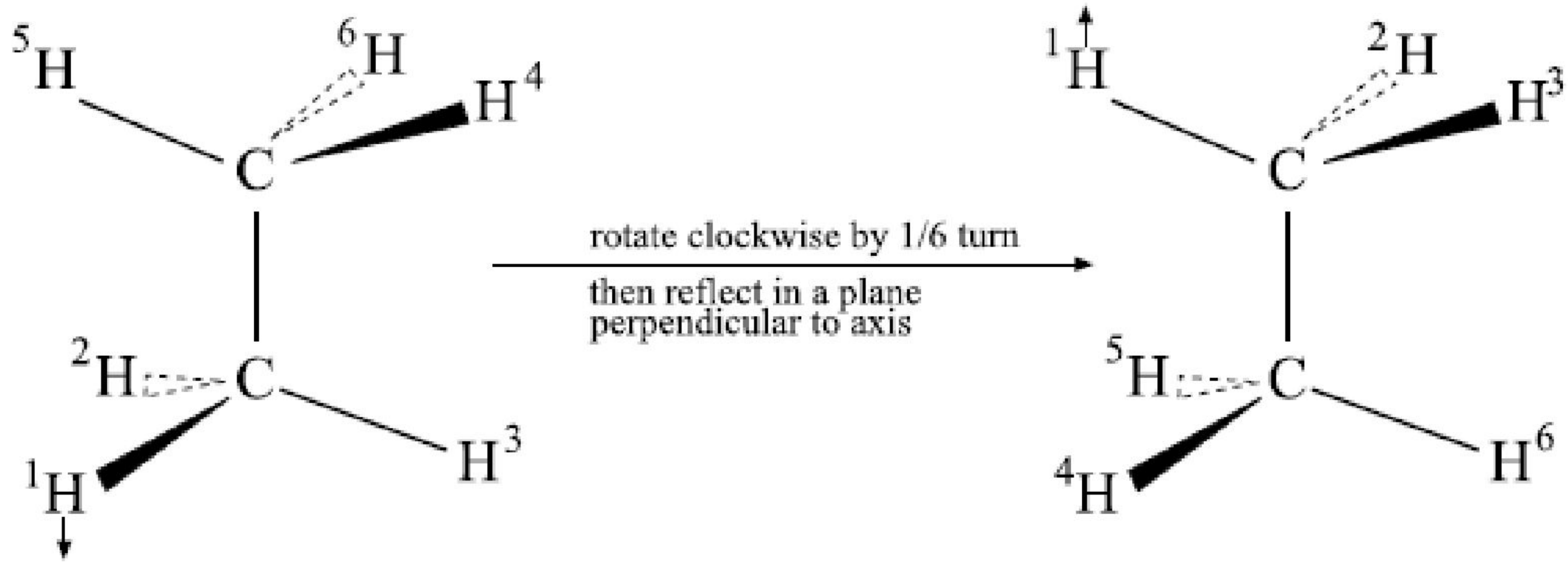



Cl
F
H
H
C
C
C
C
H
H
F
Cl

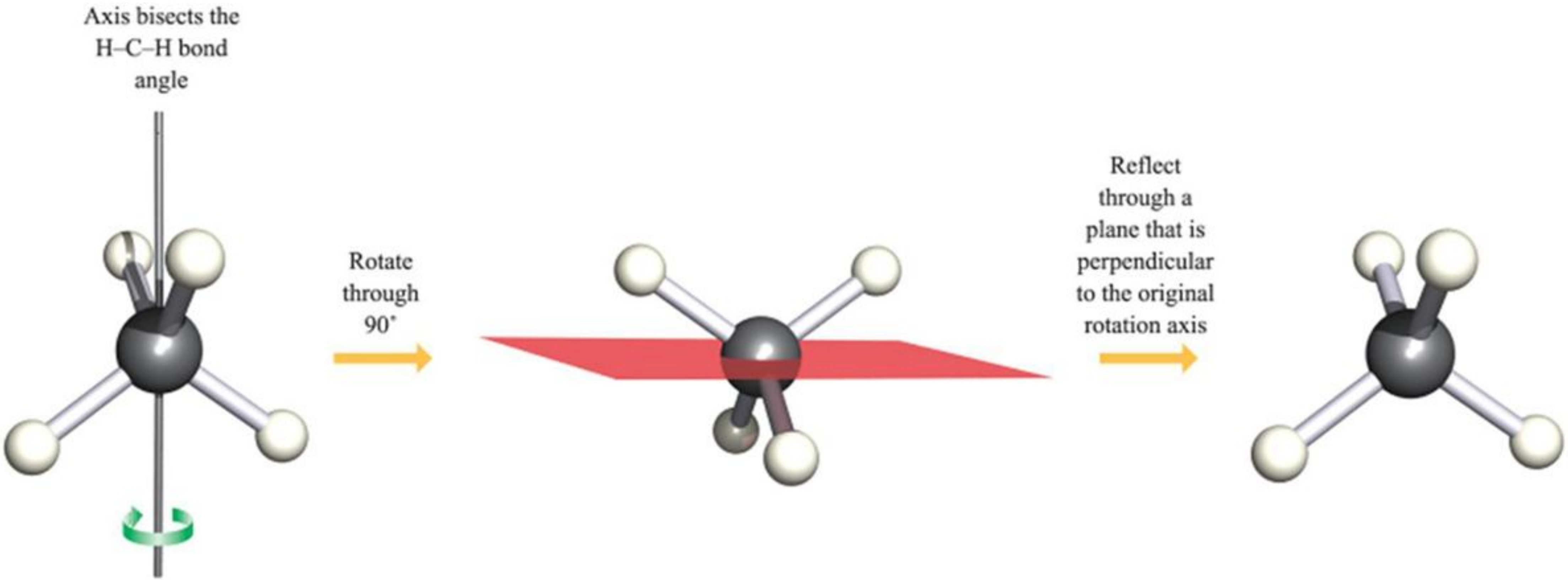
We now have the operations

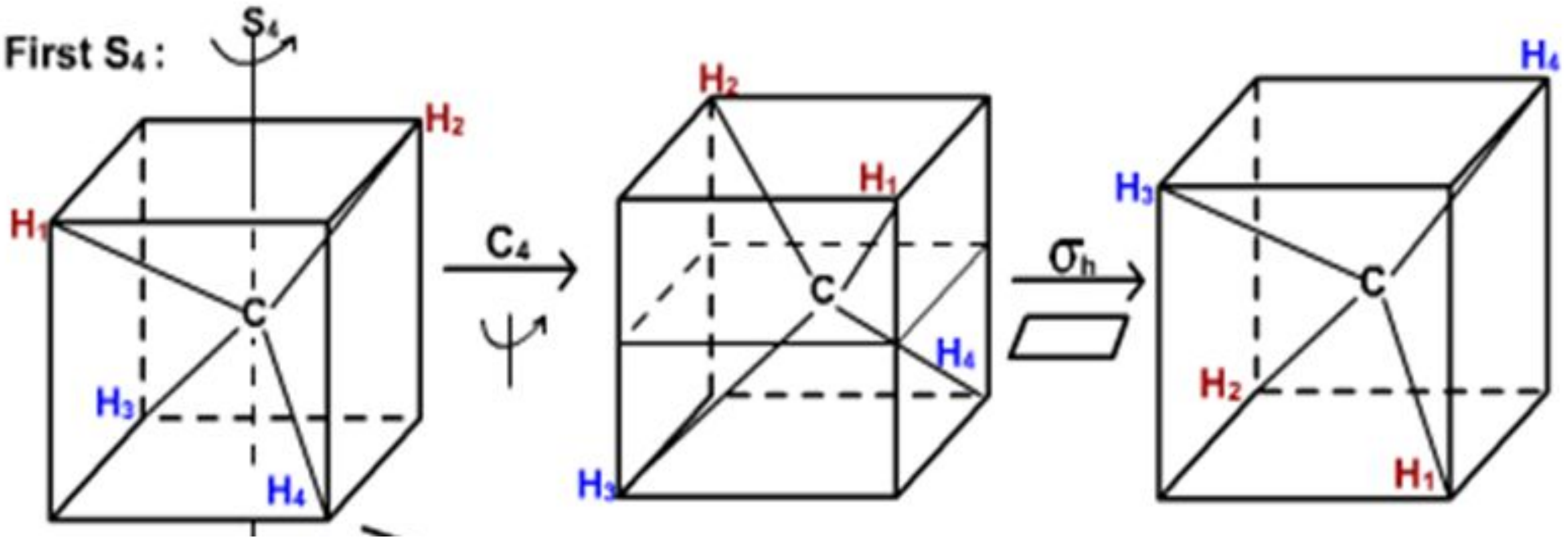
Only one more is necessary in order to specify molecular symmetry completely. That is called an IMPROPER ROTATION and is given the symbol S

The Sn operation is rotation by 1/n of a turn, followed by reflection in a plane perpendicular to the axis,



Ethane in the staggered conformation has an S6 axis because it is brought to an indistinguishable arrangement by a rotation of 1/6 of a turn, followed by reflection.


Axis bisects the
H–C–H bond
angle
Rotate
through
90°
Reflect
through a
plane that is
perpendicular
to the original
rotation axis


First S4 :
S4
H2
H1
C
H3
H4
C4
σh

Point Groups

The symmetry of a molecule can be described by listing all the symmetry elements of the molecule.
A molecule possesses a symmetry element if the application of the operation generated by the element leaves the molecule in an indistinguishable state.

Five different elements necessary to completely specify the symmetry of all possible molecules:

- **the identity**
- **proper rotation axis of order n**
- **a plane of symmetry**
- **an inversion centre**
- **improper (or rotation-reflection) axis of order n.**

It is convenient to classify all such molecules by a single symbol which summarises their symmetry.

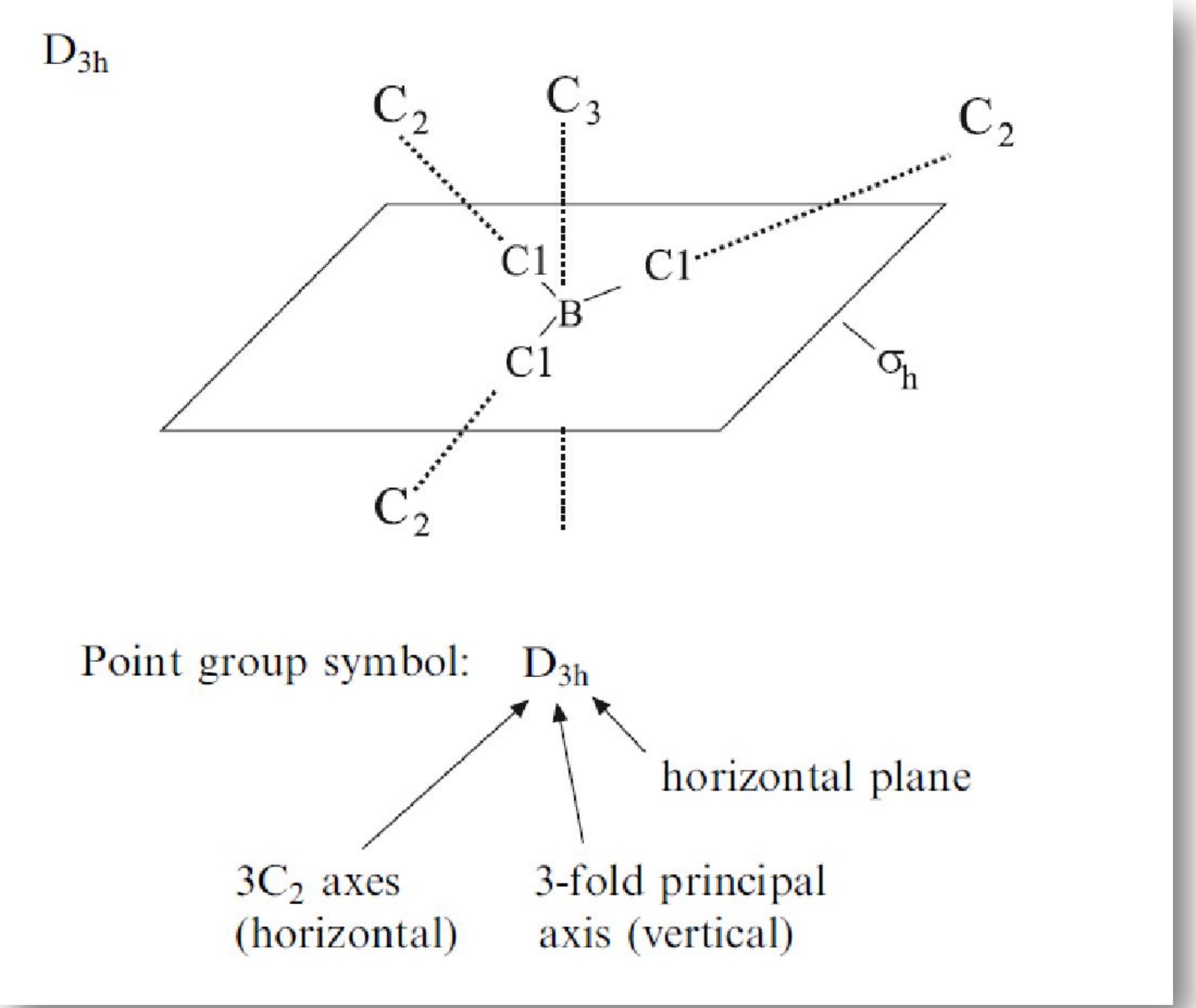
The symmetry symbol consists of three parts:

D_{3h}

The number indicates the order of the principal (i.e. highest order) axis. This is conventionally taken to be vertical.

The small letter h indicates a horizontal plane.

The capital letter D indicates that there are n(= 3 for BCl_3) C_2 axes at right angles to the principal Cn axis (C_3 for BCl_3):


D3h
C2
C3
C2
C1
C1
B
C1
σh
C2
Point group symbol: D3h
horizontal plane
3C2 axes (horizontal)
3-fold principal axis (vertical)

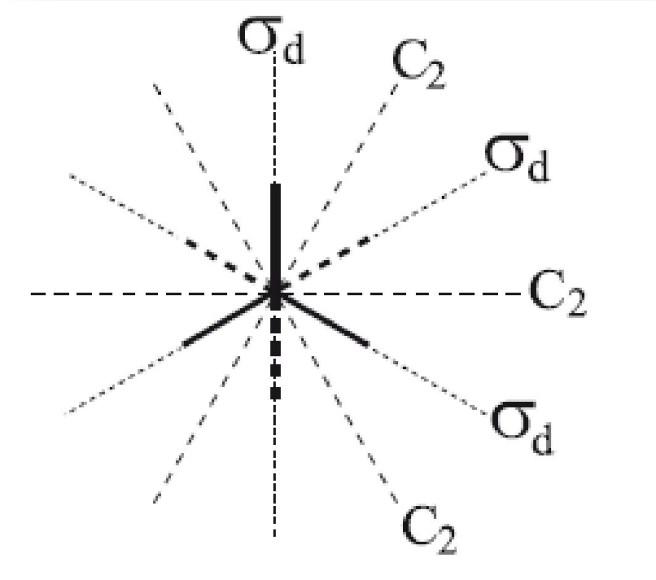
If principal axis Cn, so that its order, n, can be any number.

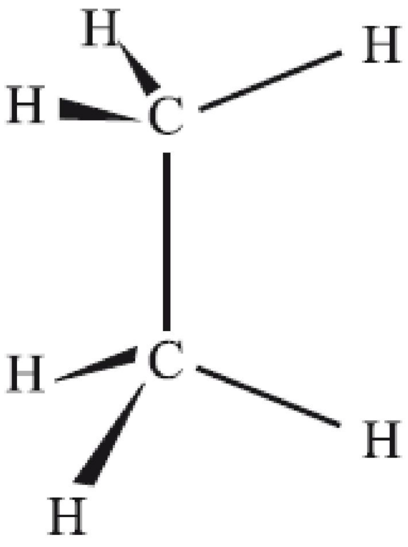
If there is no horizontal plane of symmetry,

but there are n vertical planes as well as nC_2 axes,

point group is D_{nd}.

The D and the number mean the same as before but the small d stands for DIHEDRAL PLANES, because the n vertical planes lie between the nC_2 axes.

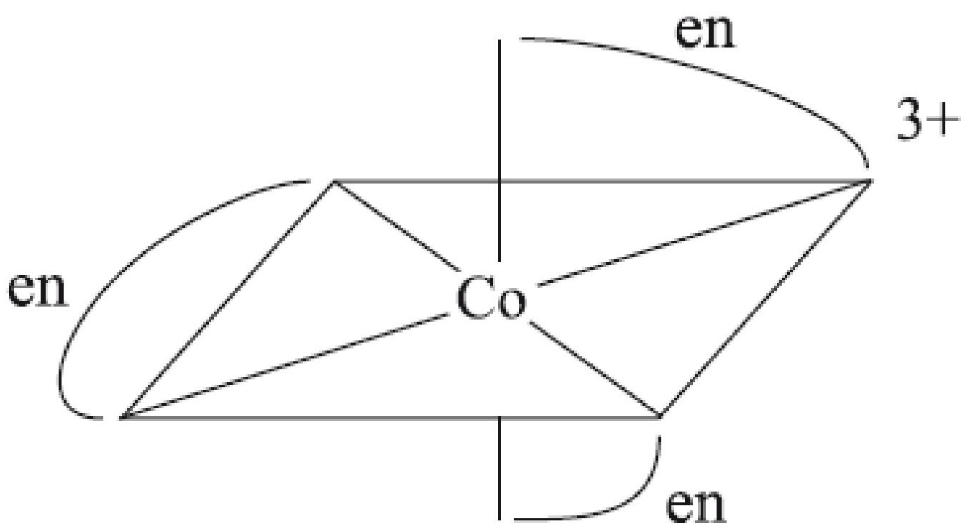

σd
C2
σd
C2
σd
C2

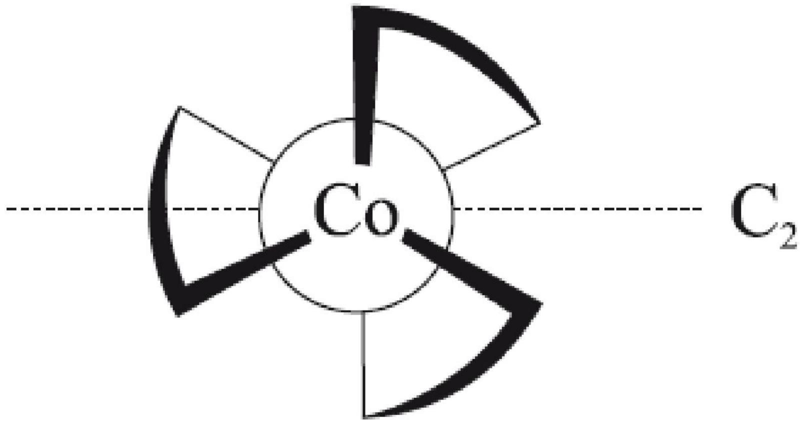


The horizontal plane takes precedence over the dihedral planes in describing the symmetry.

Some molecules have a principal Cn axis, and nC_2 axes at right angles, but no horizontal or vertical (dihedral) planes.

There is then no need to include h or d in the symmetry symbol.



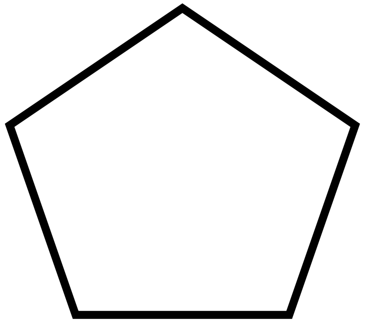


(en = $NH_2CH_2CH_2NH_2$)

If the principal C_n axis is not accompanied by nC_2 axes, the first letter of the point group is C.

A horizontal plane is looked for first, and is shown by a little h.

If is not present, n vertical planes are looked for and are shown by a small v.



Some rather rare molecules possess only two elements of symmetry, and these are given a special symbol:

E and i only Ci
E and   only Cs
E and Sn only Sn

Many molecules have no symmetry at all (i.e. their only symmetry element is the identity, E.

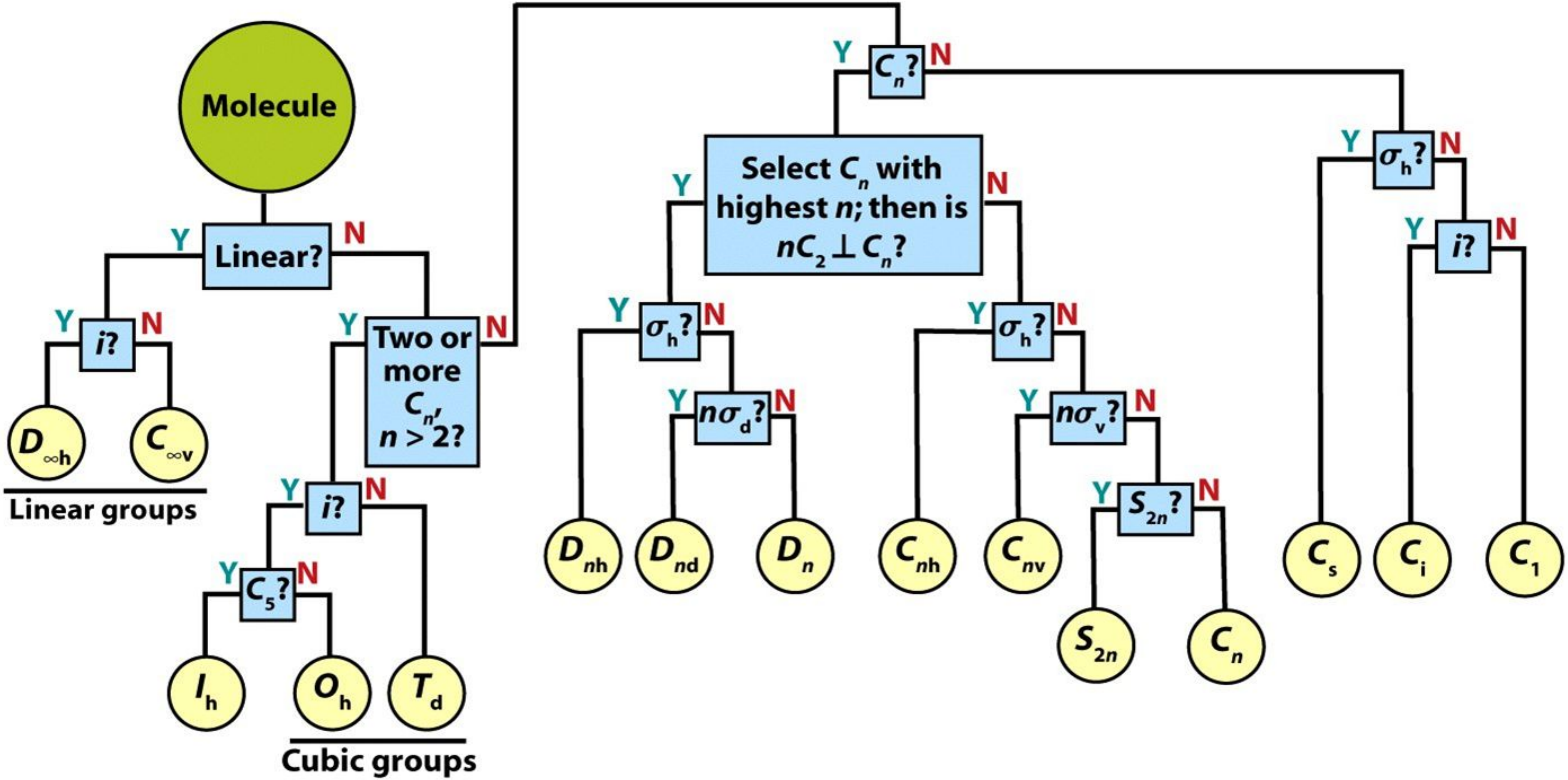


Figure 7-9
Shriver & Atkins Inorganic Chemistry, Fourth Edition